Student Handout

Name: I Forgot Again Team **Date:** 4/15/24 **Class:** Engineering

MyDesign Portfolio

Element H: Prototype Testing and Data Collection Plan

Testing Details *(H1, H2, H3, H4)*

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	The prototype must have a proper fit into the car console.	Materials: Ruler, car, car console, and prototype. 1. Measure the length, width, and height of the prototype with the ruler. 2. Measure the length, width, and height of the car console with the ruler. 3. Place the prototype into the car console. 4. Record if the device fits in the car console and if the lid iteration can open.	1	Even with an iteration, the prototype does not fit. The prototype does not fit but it can fit with an iteration. The prototype successfully fits into the car console.	Pass: The prototype successfully fits into the car console. Fail: The prototype fails to fit into the car console.
2	The prototype must keep the items intact and undamaged while in the device.	Materials: Car keys, phone, prototype, and car (NOTE: Not all cars can be stored in the device as they are required to be in the key holder of the vehicle). 1. Place both the phone and car keys into the device and close it. 2. Go on a 30-minute drive (can be any route). 3. Once stopped, record if the condition of the items.	1	The prototype and the items placed are both dented or damaged in some way. The prototype is dented or damaged in some way but the items inside are intact. The prototype is	Pass: Both the prototype and the items in the prototype are undamaged and intact. Fail: The prototype and the items placed are both dented or damaged in some

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				intact but the items inside are dented or damaged in some way. Both the prototype and the items in the prototype are undamaged and intact.	way. The prototype is dented or damaged in some way but the items inside are intact. The prototype is intact but the items inside are dented or damaged in some way.
3	The prototype must successfully hold items so that the driver can focus their full attention on the road.	Materials: Car, car keys, phone, and prototype. 1. Place both the phone and car keys into the device and close it. 2. Go on a 30-minute drive (can be any route). 3. After the drive, record if the driver is distracted while driving.	1	The driver is constantly worrying about their items and is driving distracted. The driver sometimes worries about the items and is driving distracted. The driver only worries about the items a normal amount and drives undistracted. The driver barely has to worry about their items and drives undistracted.	Pass: The driver only worries about the items a normal amount and drives undistracted. The driver barely has to worry about their items and drives undistracted. Fail: The driver is constantly worrying about their items and is driving distracted. The driver sometimes worries about the items and

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					is driving distracted.

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Feedback From Field Experts *(H5)*

Revised Testing Details That Incorporate Field Expert Feedback *(H5)*

Ryan Butchko – Works with cleaning cars and is familiar with all shapes and sizes of the interior of vehicles. Ryan reviewed our testing plan, specifically on how the prototype would fit in the console. Ryan's expertise with the fittings of a car helped give us insightful feedback on how we could scale the prototype so it could appropriately fit the middle of the car. Ryan recommended that we do ____

Ana Maria – Has a master's degree in psychology and specializes in studying behavioral disengagement. Ana informed us that "multitasking while driving is a dangerous practice, and the existing pressure of having to send that 'last text' or like that 'last picture' before hitting the gas pedal can often lead to people cracking under the pressure and ending up in an accident." So, we created our prototype so that people could place their phones in the box BEFORE they begin driving so that they aren't tempted to retrieve the device as they're driving.

All changes are in **BOLD.**

	requirement	testing setup and procedure	test cases and number of trials		expected range of results	pass/fail criteria, goal
1	The prototype must have a proper fit into the car console.	Materials: Ruler, car, car console, and prototype. 1. Measure the length, width, and height of the prototype with the ruler. 2. Measure the length, width, and height of the car console with the ruler. 3. Place the prototype into the car console. 4. Record if the **measurements of the** device fit in the car console and if the lid iteration can	1		Even with an iteration, the prototype does not fit. The prototype does not fit but it can fit with an iteration. The prototype successfully fits into the car console **and is stable**.	Pass: The prototype successfully fits into the car console **and is stable**. Fail: The prototype fails to fit into the car console.

		open.				
2	The prototype must keep the items intact and undamaged while in the device.	Materials: Car keys, phone, prototype, and car (NOTE: Not all cars can be stored in the device as they are required to be in the key holder of the vehicle). 1. Place both the phone and car keys into the device and close it. 2. Go on a 30-minute drive (can be any route). 3. Once stopped, record if the condition of the items. **4. Damage to items should be calculated using a physics gravitational equation to see how far the item has been dropped in the prototype. Force = Gravity x distance between object 1 x distance between object 2) / Distance between those objects squared.** **5. Repeat this test any amount of times in different cars for results.**	1		The prototype and the items placed are both dented or damaged in some way. The prototype is dented or damaged in some way but the items inside are intact. The prototype is intact but the items inside are dented or damaged in some way. Both the prototype and the items in the prototype are undamaged and intact.	Pass: Both the prototype and the items in the prototype are undamaged and intact. Fail: The prototype and the items placed are both dented or damaged in some way. The prototype is dented or damaged in some way but the items inside are intact. The prototype is intact but the items inside are dented or damaged in some way.
3	The prototype must	Materials: Car, car keys, phone, and prototype. (NOTE: Not all cars can be	1		The driver is constantly worrying	Pass: The driver only worries

	successfully hold items so that the driver can focus their full attention on the road.	stored in the device as they are required to be in the key holder of the vehicle). 1. Place both the phone and car keys into the device and close it. 2. Go on a 30-minute drive (can be any route). 3. After the drive, record if the driver is distracted while driving. **The driver should self-reflect and notice any changes to their emotions.** **4. Use the Driver Safety Measurement System (DSMS) Methodology to determine if the driver was distracted or not.** **5. Repeat this test any amount of times in different cars for results.**			about their items and is driving distracted. The driver sometimes worries about the items and is driving distracted. The driver only worries about the items a normal amount and drives undistracted. The driver barely has to worry about their items and drives undistracted (**The driver must be safe, even if an accident wasn't their fault).**	about the items a normal amount and drives undistracted. The driver barely has to worry about their items and drives undistracted. (**The driver must be safe, even if an accident wasn't their fault).** Fail: The driver is constantly worrying about their items and is driving distracted. The driver sometimes worries about the items and is driving distracted.

Element H: Prototype Testing and Data Collection Plan						
	5	4	3	2	1	0
1. Addressing Requirements. The final prototype Testing Plan addresses ______________ of the *high priority* design requirements.	*all or nearly all*	*many*	*some*	*a few*	*one*	*none - or is missing altogether*
2. Conduct of Tests. The Testing Plan ______________ in describing the conduct of tests (through physical and/or mathematical modeling).	*is effective*	*is generally effective*	*is only adequate*	*is partially effective*	*is at least minimally effective*	*is void*
3. Quantity of Tests Planned. The Testing Plan defines ___________________ of *all* design requirements based on the instructional context.	sufficient quantity to feasibly test *100%*	sufficient quantity to feasibly test *80%*	sufficient quantity to feasibly test *60%*	sufficient quantity to feasibly test *40%*	*sufficient quantity to feasibly test 20%*	*no feasible test*

4. Logic of the Plan. The Testing Plan provides ______________________.	a logical and *well -developed* explanation	*a logical and generally developed* explanation	*a generally logical and adequately developed* explanation	*a somewhat logical and partially-developed* explanation	*at least a generally logical and/or partially -developed explanation*	*no explanation*
5. Use of Field Experts. The Testing Plan is confirmed by _____________.	*more than two* field experts with feedback provided and incorporated	*two* field experts with feedback provided and incorporated	one field expert with feedback provided *and incorporated*	one field expert *with feedback provided*	*one field expert but no response was documented*	*Field expertise is not mentioned or addressed and field experts are not consulted.*